

ATOMIC SPECTRA

Introductory Material

Newtonian mechanics and Maxwell's electromagnetism are very successful when they are applied to macroscopic phenomena. However classical physics fails when it is applied to microscopic phenomena. For example, it predicts that the electron orbiting around the proton in a hydrogen atom will be absorbed to the proton after 1.6×10^{-11} sec by radiating electromagnetic wave, i.e. hydrogen atoms are not stable! We all know that matter is stable but classical physics can not explain this simple fact.

In 1913 Niels Bohr introduced four postulates to solve this problem:

- The electron moves around the proton in circular orbits under the influence of the Coulomb force.
- Atoms have “stationary orbits” or "stationary states" in which the electron does not radiate. Classical physics may be used to describe the motion of electrons when there are in stationary states.
- Radiation is emitted by the atom *only* when an electron *jumps* from a more energetic initial stationary state to a lower state. The frequency of the emitted light (photon) is given by the Planck-Einstein formula

$$\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda} \quad (0.1)$$

where E_i is the energy of the initial state, E_f is the energy of the final state and $E_i > E_f$.

- The electrons orbital angular momentum is quantized as

$$L = mvr = n\hbar, \quad n = 1, 2, 3, \dots \quad (0.2)$$

Using above postulates Bohr showed that, for a hydrogen-like atom with atomic number Z ,

$$E_n = -\frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} \frac{Z^2}{n^2} = -\frac{e^2}{8\pi\epsilon_0 a_0} \frac{Z^2}{n^2} \quad (0.3)$$

where

$$a_0 = \text{Bohr's radius} = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = 0.529 \times 10^{-10} m \quad (0.4)$$

By using Eq (0.1) he found

$$\frac{1}{\lambda} = \frac{m_e e^4}{4\pi\hbar^3 (4\pi\epsilon_0)^2} Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (0.5)$$

This equation was empirically found in the late 1800s and called the Rydberg-Ritz equation:

$$\frac{1}{\lambda} = R_{\infty} Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (0.6)$$

According to the Bohr's model, the Rydberg constant R_{∞} is given by

$$R_{\infty} = \frac{m_e e^4}{4\pi c h^3 (4\pi \epsilon_0)^2} = 1.09737 \times 10^7 m^{-1} = 1.09737 \times 10^{-3} \text{ \AA}^{-1} \quad (0.7)$$

Even though Bohr's model proved too simplistic for atoms with multiple electrons, it still gives good results for the Hydrogen atom, which we will consider here.

Prelaboratory Questions

1. For a Hydrogen atom, consider the transition between the $n=3$ and $n=1$ states. Calculate the photon energy, frequency, and wavelength. Identify whether the emitted photon falls into the visible, infrared, or ultraviolet region.
2. Using the Rydberg equation, determine all visible transitions in the Hydrogen atom. Describe these with their final and initial energy states, the energy of the emitted photon, as well as the frequency and wavelength of the emitted photon.

Experimental Procedure

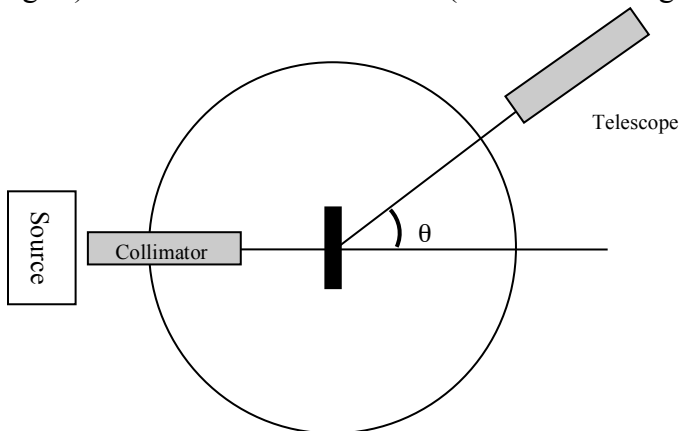
Part I : Spectral Lines of Mercury

1. Record the number of grooves/mm of your diffraction grating. Compute the separation between adjacent two lines and record it as a .
2. Find the zero order line and record the associated angle. Measure to 0.1 degree using the Vernier scale.
3. Locate and observe the three most easily visible first order lines of mercury on both sides of the zero-order line, recording the diffraction angle θ for each. Include an estimate of experimental error.
4. Calculate the (measured) wavelength λ for each of the three mercury lines, using $n\lambda = d \sin \theta$, with $n=1$.
5. Calculate the wavelength for each line from the Balmer formula or the Rydberg-Ritz formula. Compare these to your experimental results.
6. If possible, observe some second order lines ($n=2$) and use these to obtain values for the wavelengths.

Part II: Mercury Calibration Curve

In this part of experiment you will construct a graph of λ vs. $\sin\theta_m$. This graph (a calibration curve) relates a wavelength to its location of m^{th} order maximum by a diffraction grating. Once this graph is obtained, you can use it for different sources as long as the same diffraction grating is used.

You will use prominent lines in the spectra of mercury to construct a calibration curve. You will measure the locations (angles) of the line of known colors (known wavelengths) to both left and right directions.



Mercury spectrum:

Violet	404.7, 407.8 nm
Violet-blue	435.8 nm
Blue-green	491.6 nm
Green	546.1 nm
Yellow	577.0, 579.1 nm

Part III: Hydrogen/Helium/Argon/Neon Spectra

You will measure the locations (angles) of the line of known colors (but unknown wavelengths). You determine their wavelength by using the mercury calibration curve you made in Part II. You can find the Rydberg constant R_∞ from the slope of $1/\lambda$ vs. $1/n_i^2$.

Analysis Questions

1. Calculate the relative error compared with the standard value for each of the mercury wavelengths.
2. Compare your result with the accepted value: $1.097 \times 10^{-3} \text{ \AA}^{-1}$. In other words, compute the percentage difference.
3. Compare the slope of the mercury calibration plot λ vs. $\sin\theta_m$ with the actual slit width a .